SD card Partition

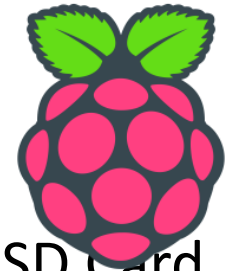
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What can You Learn?

- 建置embedded system: Raspberry Pi + Linux
- 更細部的客製化自己的embedded system
  - 透過 fdisk 調整SD card中 root filesystem的 partition大小

Booting Sequence in Pi (1)



- Execute the boot ROM (the first stage bootloader)
  - This firmware mainly enables the GPU to access the SD Card, and reads a FAT32 partition on it.
  - The FAT filesystem reading code was implemented in the boot ROM to read files (Pi firmware).
- Next the boot ROM checks each of the boot sources for a file called **bootcode.bin** (the second stage bootloader).
  - The GPU reads this file and executes it.
  - If it is successful, it will load the code into the local 128K cache and jump to it.
- **bootcode.bin** in turn then loads and runs **start.elf** (and **fixup.dat**) also from the first partition.
- **start.elf** then reads **config.txt** and sets up any GPU configuration requested.

config.txt

```
start_file=start.elf
fixup_file=fixup.dat

kernel=Image

# To use an external initramfs file
#initramfs rootfs.cpio.gz

# Disable overscan assuming the display supports displaying the full resolution
# If the text shown on the screen disappears off the edge, comment this out
disable_overscan=1

# How much memory in MB to assign to the GPU on Pi models having
# 256, 512 or 1024 MB total memory
gpu_mem_256=100
gpu_mem_512=100
gpu_mem_1024=100

# Enable UART0 for serial console on ttyAMA0
dtoverlay=miniuart-bt

# enable 64bits support
arm_64bit=1
```

- 32-bit Linux: zImage
- 64-bit Linux: Image

Booting Sequence in Pi (2)

- **start.elf** reads **cmdline.txt** and loads and runs Linux kernel image, Image, passing it the entire command-line that it read from **cmdline.txt**.

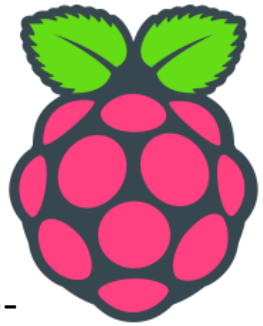
There is a **`kernel = Image`** option in **config.txt**.

- **Linux Kernel**
  - Decompress the kernel into SDRAM.
  - Setup peripherals, such as LCD, HDMI, I2C, USB, audio, … etc.
  - Mount the Linux root filesystem that contains all userspace libraries and applications.

The root filesystem is on the second partition, so there'll be a **`root=/dev/mmcblk0p2`** option somewhere in **cmdline.txt**.

Booting Sequence in Pi (3)

- The Linux kernel then mounts the **`/dev/mmcblk0p2`** partition as the root (/) filesystem, and continues booting the rest of the system from there.
- In some cases the **`/etc/fstab`** file (File System TABle) on the root filesystem will have a line asking for the **`/dev/mmcblk0p1`** partition to be mounted at **`/boot`**.
  - Easy to modify **config.txt** and **cmdline.txt**

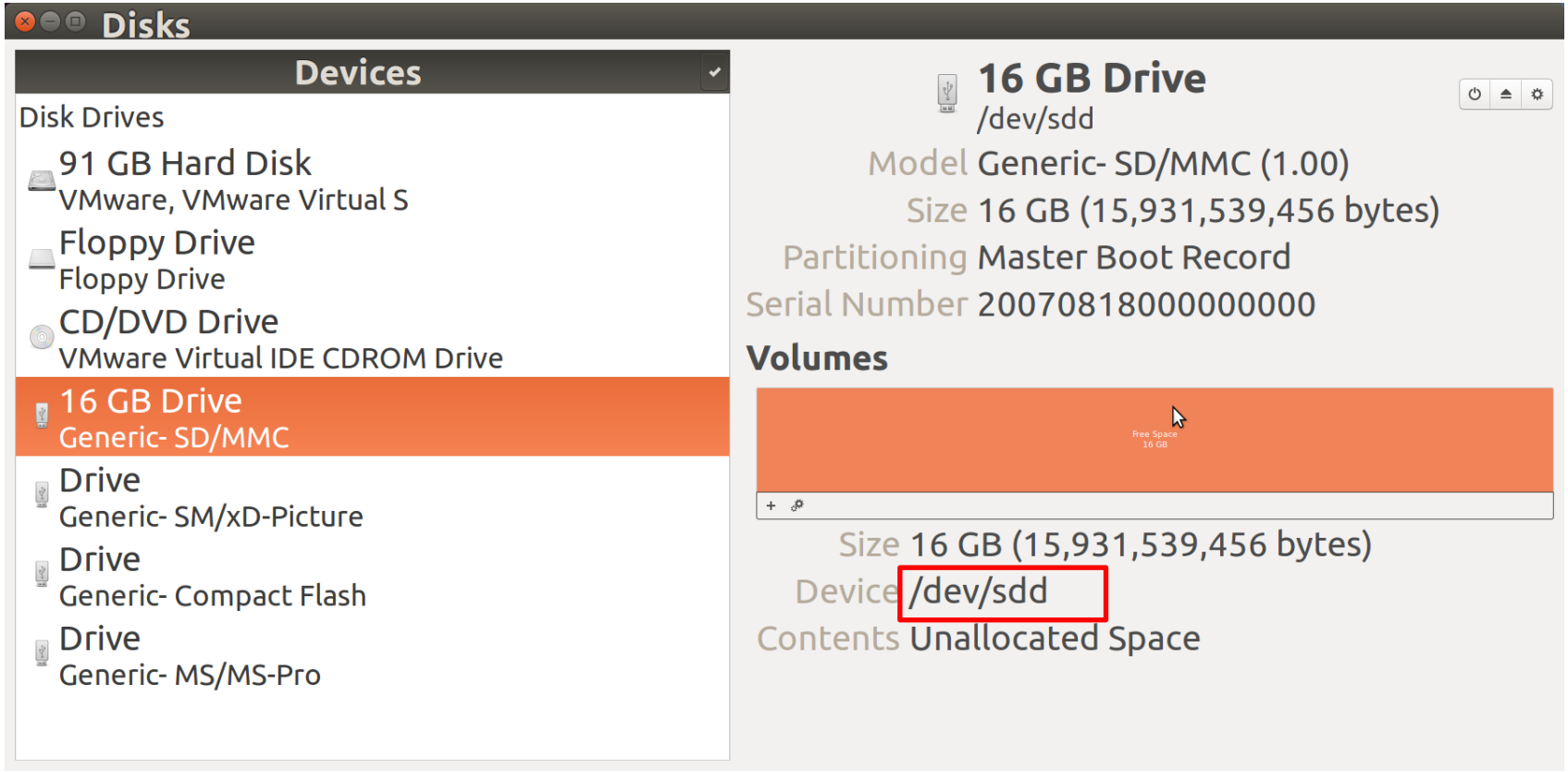


Reference from https://github.com/raspberrypi/noobs/wiki/Standalone-partitioning-explained

Partition SD Card

- Insert your SD card into your Linux box
- Do not mount it
- 確認/dev裡哪一個檔案代表你的SD card
  - 可使用Ubuntu裡的disk utility or disk顯示相關訊息
    - `lsblk`
    - `dmesg`
  - 在稍後的操作範例裡，我們假設是**/dev/sdd**

Disks in Ubuntu



Partition SD Card Using fdisk

- **SD card裡的資料將被清除**，若有重要資料，請事先備份
- 將SD card插入讀卡機，並與電腦相連，並確認SD card尚未被partition
- 這裡，SD card讀卡機的 device file是/dev/sdd

$ sudo -s

$ fdisk /dev/sdd

(目前SD card包含兩個partition: sdd1, sdd2)

```
root@ubuntu: ~
root@ubuntu:~# fdisk /dev/sdd

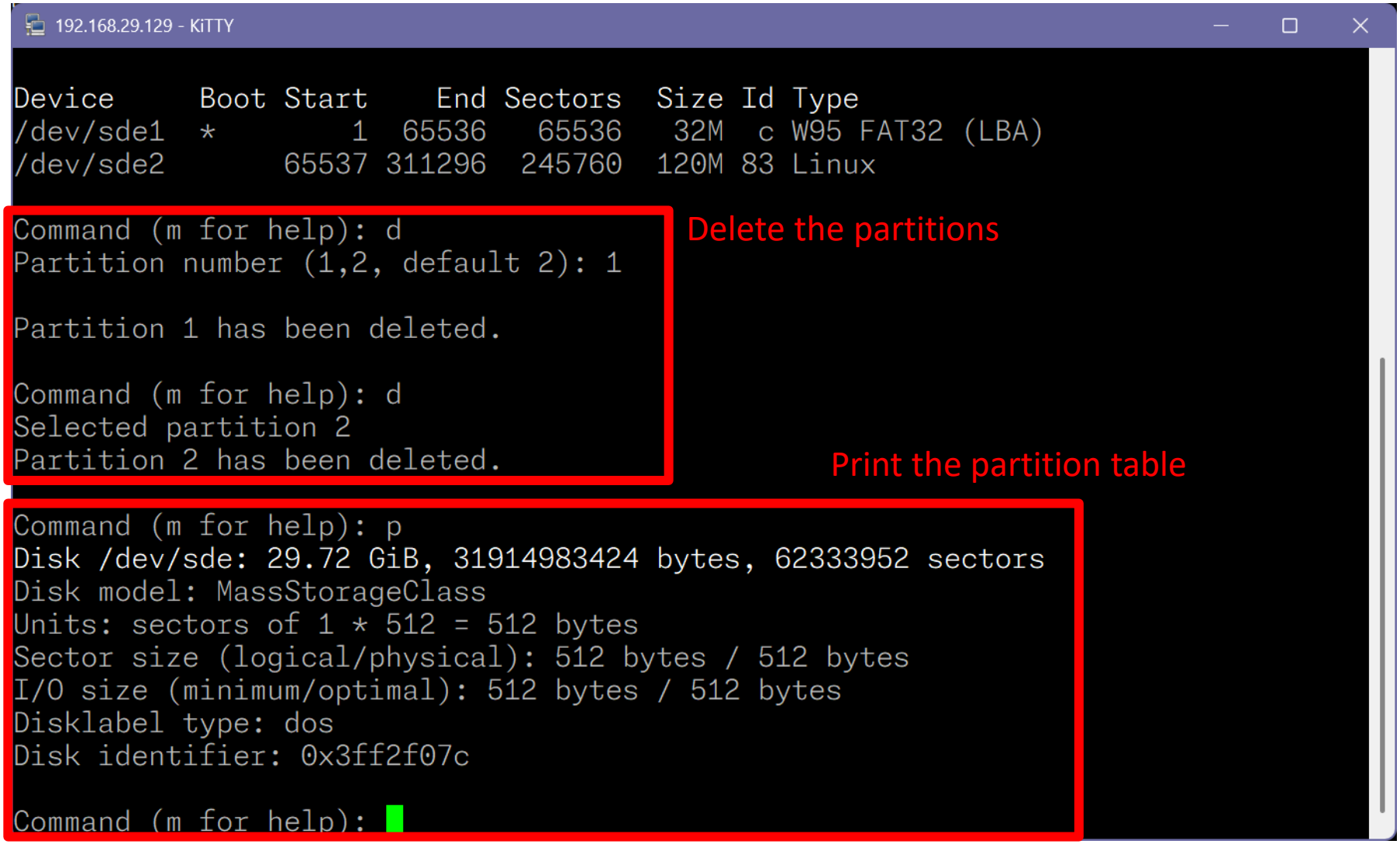
Command (m for help): p

Disk /dev/sdd: 15.9 GB, 15931539456 bytes
64 heads, 32 sectors/track, 15193 cylinders, total 31116288 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk identifier: 0x5b155e53

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1   *           1       65536       32768    c  W95 FAT32 (LBA)
/dev/sdd2           65537    16842752     8388608   83  Linux

Command (m for help):
```

Delete All the Partitions



Partition SD Card Using fdisk

- 若Disklabel type為sun，需先調整回DOS

```
Command (m for help): p

Disk /dev/sdd: 14.86 GiB, 15931539456 bytes, 31116288 sectors
Disk model: STORAGE DEVICE
Geometry: 64 heads, 32 sectors/track, 15193 cylinders
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: sun

Device        Start      End  Sectors  Size Id Type         Flags
/dev/sdd1         0 31012863 31012864 14.8G 83 Linux native
/dev/sdd2  31012864 31115263   102400   50M 82 Linux swap      u
/dev/sdd3         0 31115263 31115264 14.9G  5 Whole disk

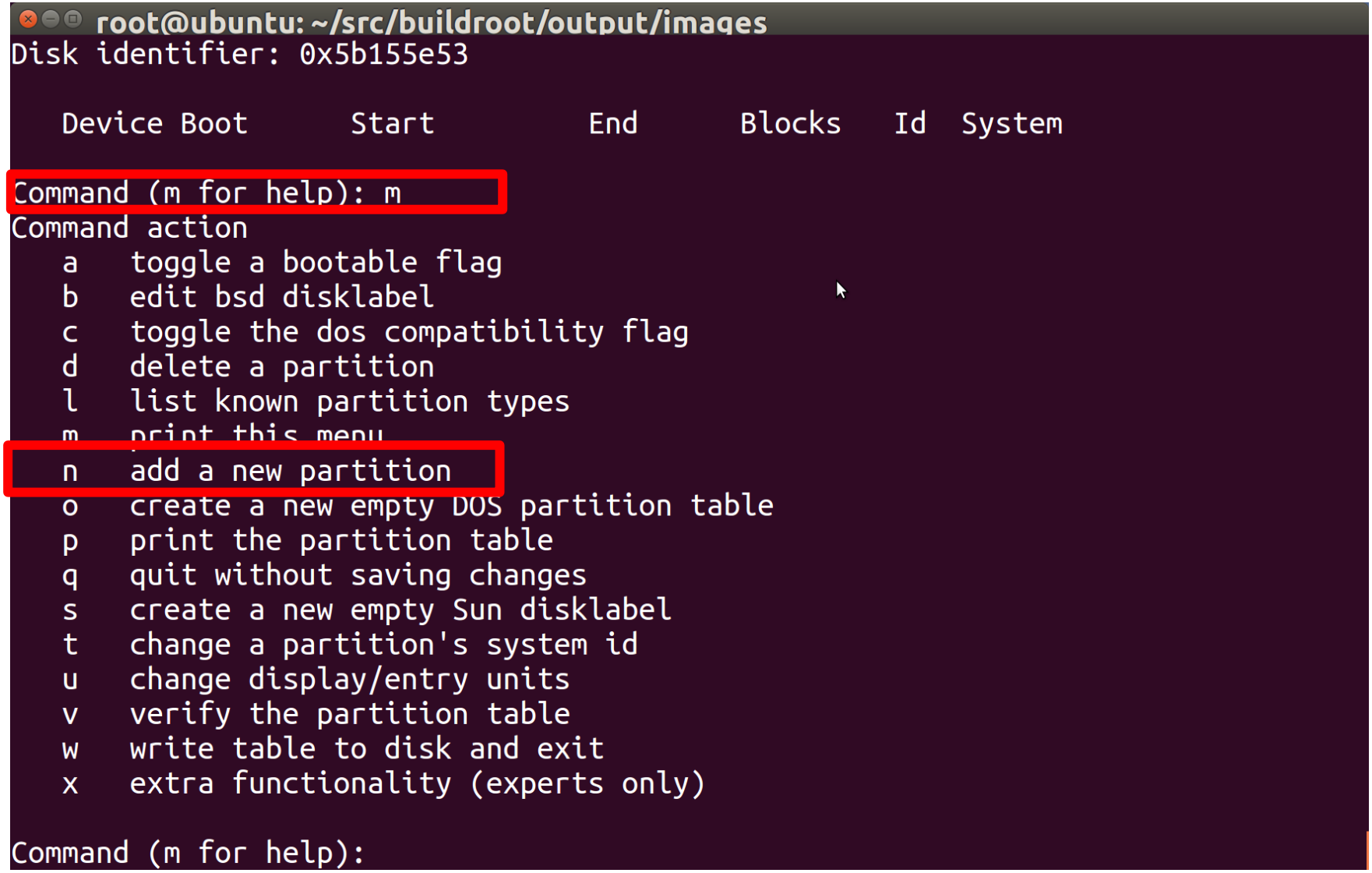
Command (m for help): o
Created a new DOS disklabel with disk identifier 0xa940e9b0.
```

```
Create a new label
 g   create a new empty GPT partition table
 G   create a new empty SGI (IRIX) partition table
 o   create a new empty MBR (DOS) partition table
 s   create a new empty Sun partition table
```

Partition Arrangement

- **FAT32 (primary partition)**
  - Size: 64MB
  - Bootable
- **Linux ext4 (primary partition)**
  - Size: the rest of space.

    (假設SD card是16GB)


root@ubuntu: ~/src/buildroot/output/images
Disk identifier: 0x5b155e53
Device Boot Start End Blocks Id System
Command (m for help): m
Command action
a toggle a bootable flag
b edit bsd disklabel
c toggle the dos compatibility flag
d delete a partition
l list known partition types
m print this menu
n add a new partition
o create a new empty DOS partition table
p print the partition table
q quit without saving changes
s create a new empty Sun disklabel
t change a partition's system id
u change display/entry units
v verify the partition table
w write table to disk and exit
x extra functionality (experts only)
Command (m for help):

Create Partition: FAT32

```
root@ubuntu: ~/src/buildroot/output/images
root@ubuntu:~/src/buildroot/output/images# fdisk /dev/sdd

Command (m for help): p

Disk /dev/sdd: 15.9 GB, 15931539456 bytes
64 heads, 32 sectors/track, 15193 cylinders, total 31116288 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk identifier: 0xbaf7cf51

   Device Boot      Start         End      Blocks   Id  System

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-31116287, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-31116287, default 31116287): +64M

Command (m for help):
```

Create Partition: Linux ext4

```
Command (m for help): n
Partition type:
   p   primary (1 primary, 0 extended, 3 free)
   e   extended
Select (default p): p
Partition number (1-4, default 2): 2
First sector (133120-31116287, default 133120):
Using default value 133120
Last sector, +sectors or +size{K,M,G} (133120-31116287, default 31116287):
Using default value 31116287

Command (m for help):
```

檢視目前分割狀態

16GB microSD

```
Command (m for help): p

Disk /dev/sdd: 15.9 GB, 15931539456 bytes
64 heads, 32 sectors/track, 15193 cylinders, total 31116288 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk identifier: 0xbaf7cf51

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1            2048      133119       65536   83  Linux
/dev/sdd2          133120    31116287    15491584   83  Linux

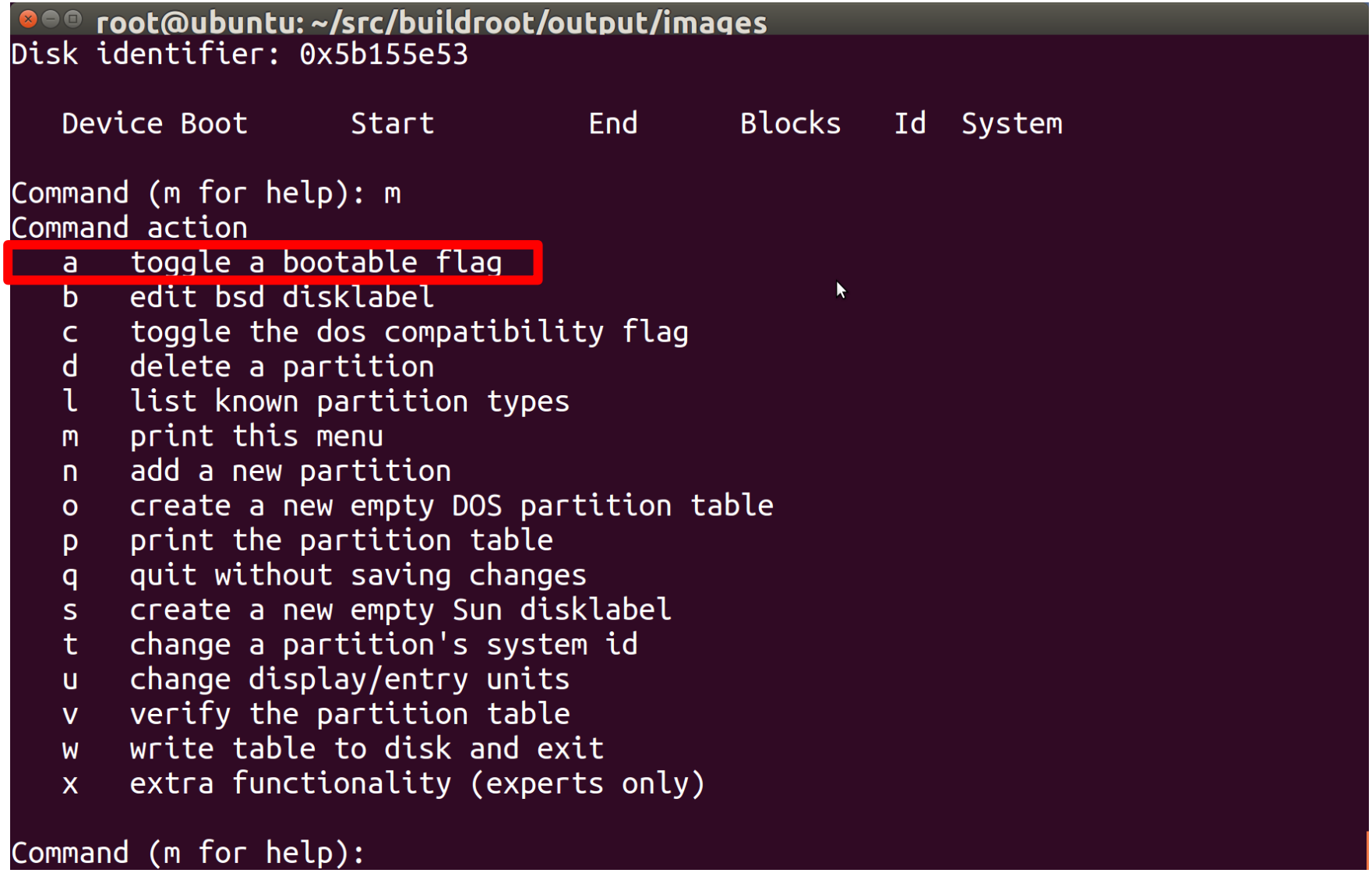
Command (m for help):
```

檢視目前分割狀態

32GB microSD

```
Command (m for help): p
Disk /dev/sde: 29.72 GiB, 31914983424 bytes, 62333952 sectors
Disk model: MassStorageClass
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x3ff2f07c

Device     Boot  Start      End  Sectors  Size Id Type
/dev/sde1         2048   133119   131072   64M 83 Linux
/dev/sde2       133120 62333951 62200832 29.7G 83 Linux
```


root@ubuntu: ~/src/buildroot/output/images
Disk identifier: 0x5b155e53
Device Boot Start End Blocks Id System
Command (m for help): m
Command action
a toggle a bootable flag
b edit bsd disklabel
c toggle the dos compatibility flag
d delete a partition
l list known partition types
m print this menu
n add a new partition
o create a new empty DOS partition table
p print the partition table
q quit without saving changes
s create a new empty Sun disklabel
t change a partition's system id
u change display/entry units
v verify the partition table
w write table to disk and exit
x extra functionality (experts only)
Command (m for help):

Mark the Partition 1 Bootable

```
root@ubuntu: ~/src/buildroot/output/images
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk identifier: 0xbaf7cf51

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1            2048      133119       65536   83  Linux
/dev/sdd2          133120    31116287    15491584   83  Linux

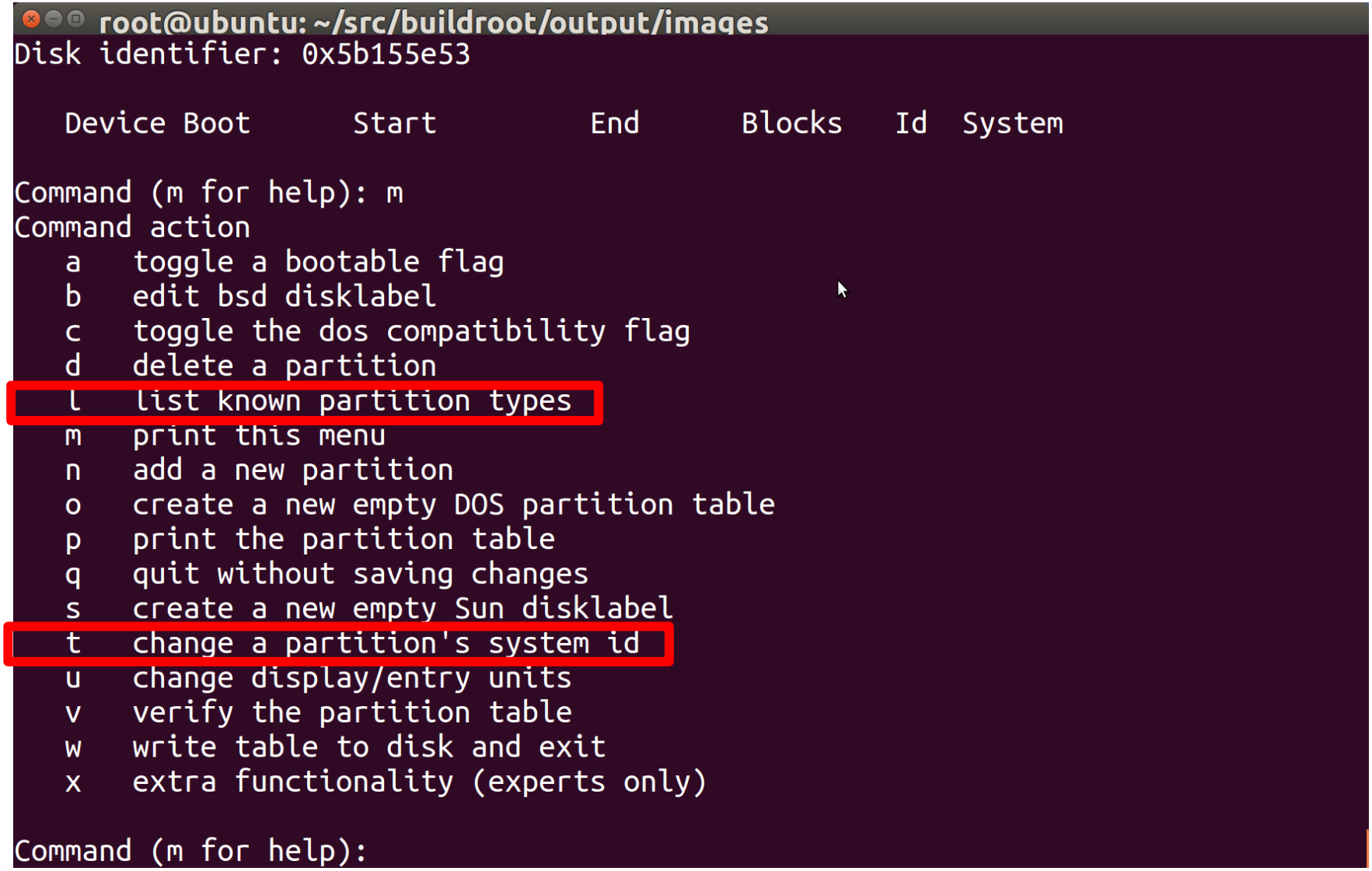
Command (m for help): a
Partition number (1-4): 1

Command (m for help): p

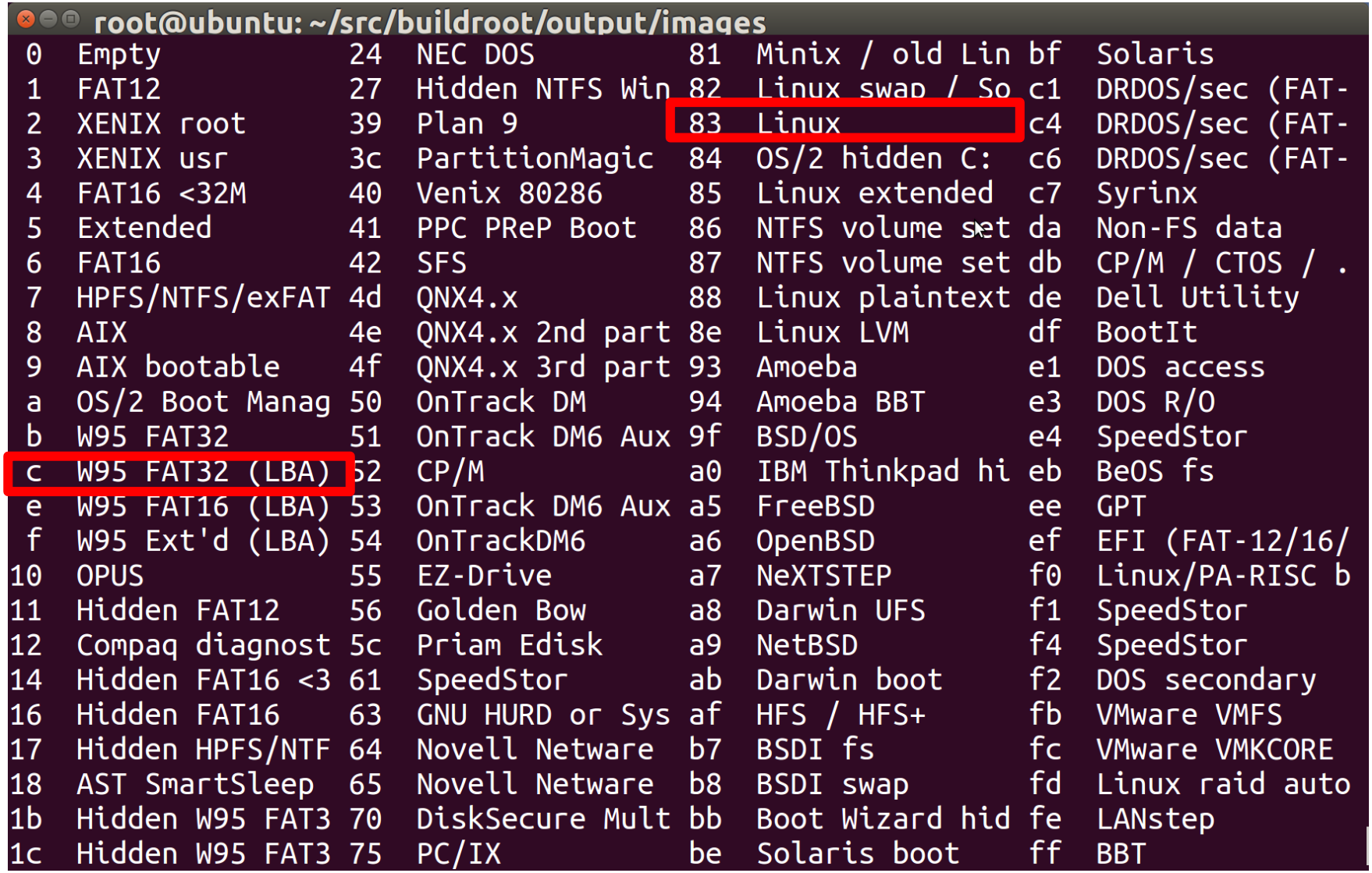
Disk /dev/sdd: 15.9 GB, 15931539456 bytes
64 heads, 32 sectors/track, 15193 cylinders, total 31116288 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk identifier: 0xbaf7cf51

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1   *        2048      133119       65536   83  Linux
/dev/sdd2          133120    31116287    15491584   83  Linux

Command (m for help):
```


root@ubuntu: ~/src/buildroot/output/images
Disk identifier: 0x5b155e53
Device Boot Start End Blocks Id System
Command (m for help): m
Command action
a toggle a bootable flag
b edit bsd disklabel
c toggle the dos compatibility flag
d delete a partition
l list known partition types
m print this menu
n add a new partition
o create a new empty DOS partition table
p print the partition table
q quit without saving changes
s create a new empty Sun disklabel
t change a partition's system id
u change display/entry units
v verify the partition table
w write table to disk and exit
x extra functionality (experts only)
Command (m for help):

Partition Types



Change Partition Type of the Partition 1

```
root@ubuntu: ~/src/buildroot/output/images

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1   *        2048      133119       65536   83  Linux
/dev/sdd2          133120    31116287    15491584   83  Linux

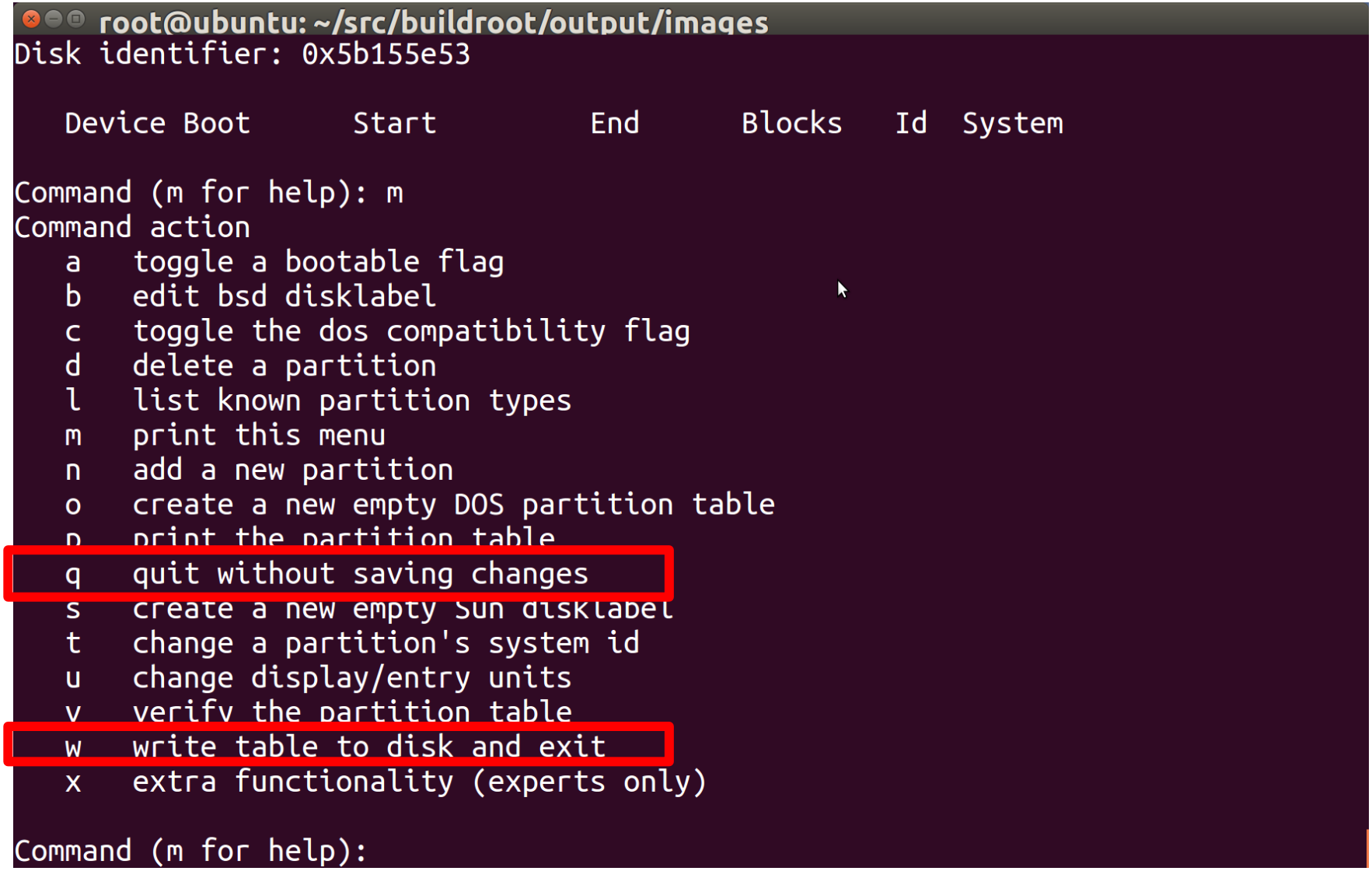
Command (m for help): t
Partition number (1-4): 1
Hex code (type L to list codes): c
Changed system type of partition 1 to c (W95 FAT32 (LBA))

Command (m for help): p

Disk /dev/sdd: 15.9 GB, 15931539456 bytes
64 heads, 32 sectors/track, 15193 cylinders, total 31116288 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk identifier: 0xbaf7cf51

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1   *        2048      133119       65536    c  W95 FAT32 (LBA)
/dev/sdd2          133120    31116287    15491584   83  Linux

Command (m for help):
```


root@ubuntu: ~/src/buildroot/output/images
Disk identifier: 0x5b155e53
Device Boot Start End Blocks Id System
Command (m for help): m
Command action
a toggle a bootable flag
b edit bsd disklabel
c toggle the dos compatibility flag
d delete a partition
l list known partition types
m print this menu
n add a new partition
o create a new empty DOS partition table
p print the partition table
q quit without saving changes
s create a new empty Sun disklabel
t change a partition's system id
u change display/entry units
v verify the partition table
w write table to disk and exit
x extra functionality (experts only)
Command (m for help):

Write the Partition Table

- “w” => write the partition table
- “q” => 若過程中有打錯，隨時放棄重來

```
Command (m for help): w
The partition table has been altered!

Calling ioctl() to re-read partition table.

WARNING: If you have created or modified any DOS 6.x
partitions, please see the fdisk manual page for additional
information.
Syncing disks.
root@ubuntu:~/src/buildroot/output/images#
```

Format Partitions

```
//Change to root user
$ sudo -s

$ mkfs.vfat -n BOOT /dev/sdd1 (以sdd為例)
   $ sudo -s
   $ apt install dosfstools

$ mkfs.ext4 -L filesystem /dev/sdd2 (以sdd為例)
```

```
root@ubuntu:~/src/buildroot/output/images# mkfs.vfat -n BOOT /dev/sdd1
mkfs.fat 3.0.26 (2014-03-07)
root@ubuntu:~/src/buildroot/output/images#
root@ubuntu:~/src/buildroot/output/images# mkfs.ext4 -L filesystem /dev/sdd2
mke2fs 1.42.9 (4-Feb-2014)
Filesystem label=filesystem
OS type: Linux
Block size=4096 (log=2)
Fragment size=4096 (log=2)
Stride=0 blocks, Stripe width=0 blocks
969136 inodes, 3872896 blocks
193644 blocks (5.00%) reserved for the super user
First data block=0
Maximum filesystem blocks=3967811584
119 block groups
32768 blocks per group, 32768 fragments per group
8144 inodes per group
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done
```

32GB microSD

192.168.29.129 - KiTTY

```
root@smallturtlenb6:/home/pschen# mkfs.vfat -n BOOT /dev/sde1
mkfs.fat 4.2 (2021-01-31)
root@smallturtlenb6:/home/pschen#
root@smallturtlenb6:/home/pschen# mkfs.ext4 -L filesystem /dev/sde2
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 7775104 4k blocks and 1945888 inodes
Filesystem UUID: ff6ee4b1-43c8-47ea-bce3-8d6651c61879
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
        4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks):
done
Writing superblocks and filesystem accounting information: done

root@smallturtlenb6:/home/pschen#
root@smallturtlenb6:/home/pschen#
```

Mount Partitions

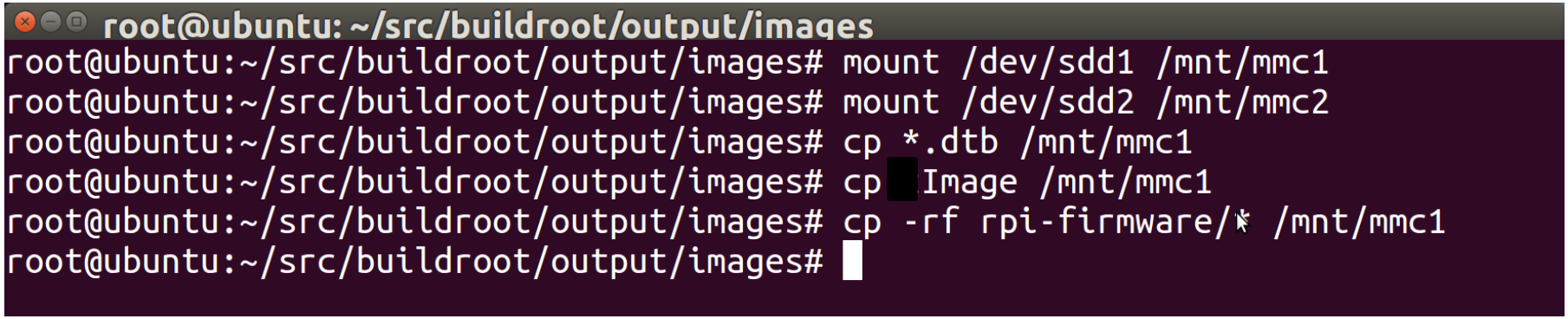
```
//Change to root user
$ sudo –s

$ mkdir /mnt/mmc1

$ mount /dev/sdd1 /mnt/mmc1
```

Copy Files for the Partition BOOT

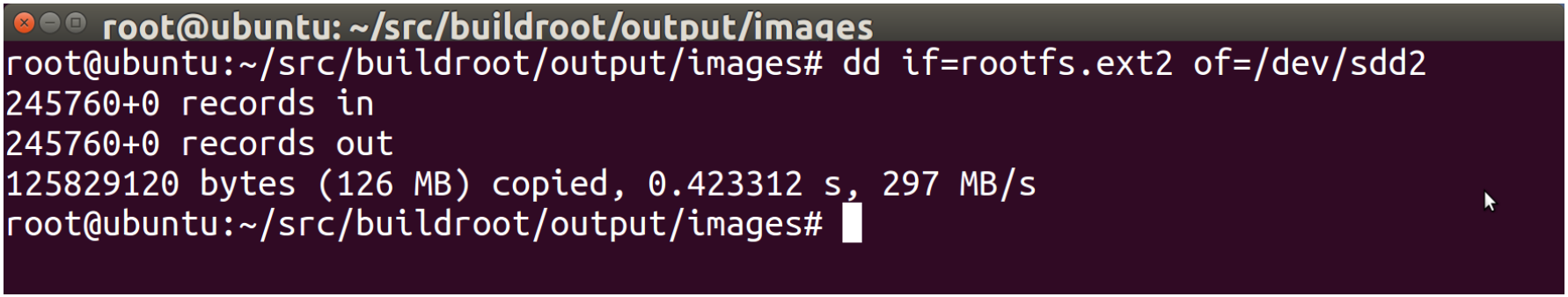
```
$ cd <buildroot目錄>/output/images/
$ cp *.dtb /mnt/mmc1
$ cp Image /mnt/mmc1
$ cp -rf rpi-firmware/* /mnt/mmc1
```



Copy Files for the Partition filesystem

```
$ cd <buildroot目錄>/output/images/

// rootfs.ext2是image file
$ dd if=rootfs.ext2 of=/dev/sdd2
```



Testing

- Change to root

```
$ sudo -s
```

- Umount SD card

```
$ umount /dev/sdd1
```

- 取出SD card，放入Raspberry Pi，重新開機

Q&A

- 請解釋下列指令、參數的意義

```
$ mkfs.ext4 -L filesystem /dev/sdd2
```